BACKEND ENGINEERING · HLD INTERVIEW PREP

Back-of-the-Envelope
Estimation — Study Notes

Traffic · Storage · RAM · App Servers
Facebook Case Study · Formulas · CAP Theorem

TOPICS COVERED

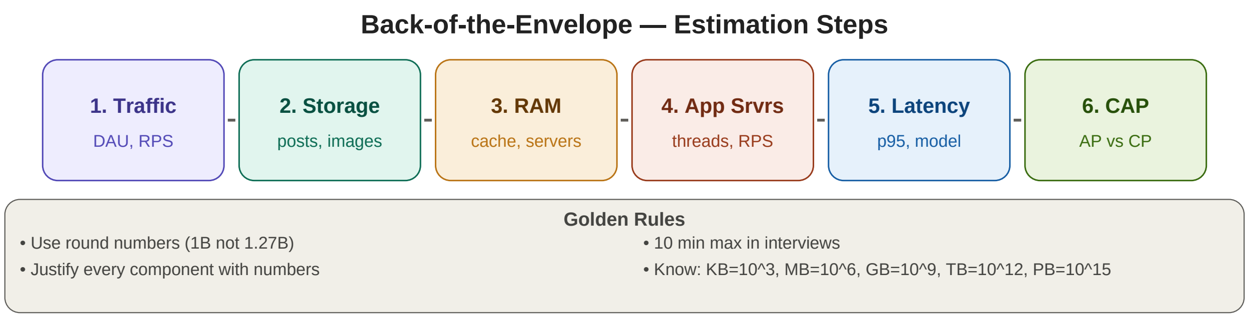
Table of Contents

1. What is Back-of-the-Envelope Estimation?

Back-of-the-Envelope (BoE) estimation is a quick, rough calculation done *before* system design to estimate key infrastructure parameters — servers, storage, RAM, bandwidth. It prevents over-engineering and gives numerical justification for every design component.

- Avoids building 1,000-server systems for a 100-user app
- Gives interviewers confidence you understand scale
- Takes 5–10 minutes max — rough numbers, not precise
- Use multiples of 10 and round aggressively (1B not 1.27B)
- Every design decision should trace back to an estimation number

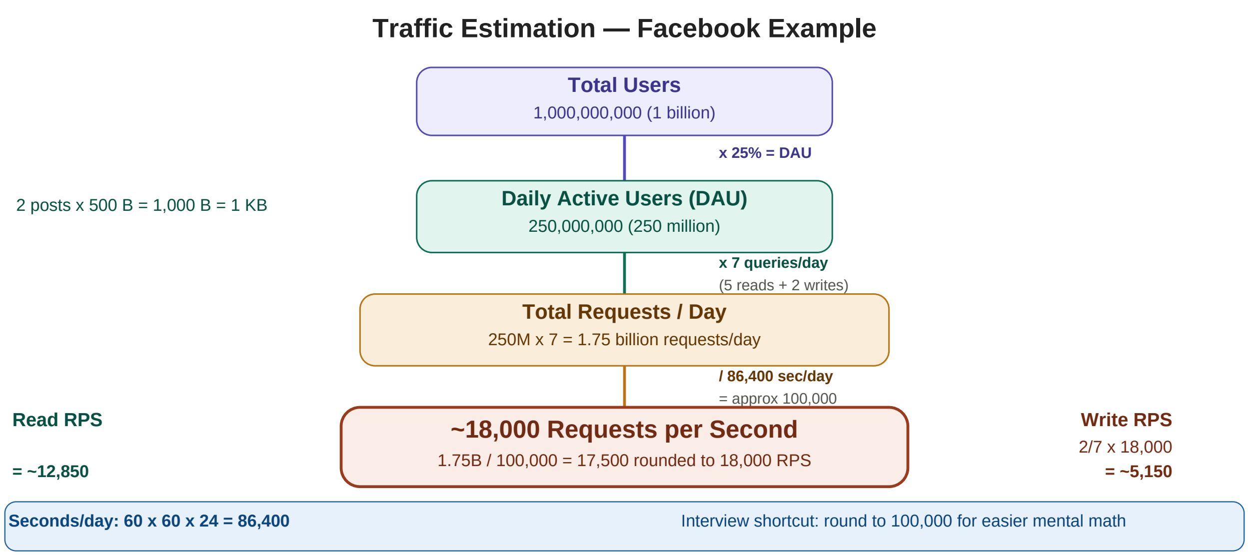
2. The 6-Step Process



6 estimation steps — always in this order · max 10 minutes in interviews

3. Step 1: Traffic Estimation (RPS)

Requests Per Second (RPS) is the most important number in traffic estimation. It determines how many servers you need and whether you need load balancing, caching, and queuing.



1B total users → 250M DAU → 1.75B requests/day → 18,000 RPS · read:write = 5:2

Formula: RPS = (DAU x queries_per_user_per_day) / 86,400

Interview shortcut: divide by 100,000 instead of 86,400 for faster mental math (conservative estimate).

Always split RPS into Read RPS and Write RPS — they have different infrastructure implications.

4. Step 2: Storage Estimation

Storage estimation covers **what data is written daily** and how that accumulates over the system lifespan (typically 5 years in interviews). Always estimate both structured data (posts, metadata) and unstructured data (images, videos) separately.

Storage Estimation — Posts + Images (5-year horizon)

Text Posts

Post size:
250 chars x 2 bytes = 500 bytes
Posts per DAU per day:
2 posts x 500 bytes = 1,000 bytes = 1 KB
Daily total:
250M x 1 KB = 250 GB/day
5-year total (approx 2,000 days):
250 GB x 2,000 days = 500 TB

Image Uploads

Upload rate:
10% of 250M DAU = 25M uploads/day
Image size:
Avg 300 KB per image
Daily total:
25M x 300 KB = 7,500 GB = ~8 TB/day
5-year total (2,000 days):
8 TB x 2,000 days = 16 PB

5-Year Storage Summary

Text posts: 500 TB
250 GB/day x 2,000 days

Images: 16 PB
8 TB/day x 2,000 days

Storage Unit Cheat Sheet

1 KB = 10^3 bytes · 1 MB = 10^6 bytes · 1 GB = 10^9 bytes · 1 TB = 10^12 bytes · 1 PB = 10^15 bytes
1 char (Unicode) = 2 bytes · 1 int = 4 bytes · 1 long/double = 8 bytes · 1 image (avg) = 200–300 KB

Daily: 250 GB (posts) + 8 TB (images) · 5-year total: 500 TB (posts) + 16 PB (images)

5. Steps 3 & 4: RAM and Application Server Estimation

RAM Server Estimation

Cache (RAM) Estimation

Cache strategy:
Keep last 5 posts per DAU in memory
Memory per user:
5 posts x 500 bytes = 2,500 bytes ~= 3 KB
Total RAM needed:
250M DAU x 3 KB = 750 GB
Number of cache servers:
750 GB / 75 GB = 10 servers

10 Cache Servers (75 GB RAM each)
Stores hot data for 250M active users

Application Server Estimation

Thread model:
1 server = 50 threads
Throughput per thread:
1 req / 500ms = 2 requests/sec per thread
Server capacity:
50 threads x 2 RPS = 100 RPS/server
Servers needed:
18,000 / 100 = 180 servers

180 Application Servers
100 RPS each, 50 threads @ 500ms latency

Full Infrastructure Summary

App Servers:	180 (100 RPS each)	**Cache Servers:**	10 (75 GB RAM each)	**Total RAM:**	750 GB
Peak RPS:	18,000	**Latency target:**	500 ms (p95)	**Storage (5yr):**	500 TB (posts) + 16 PB (images)

10 cache servers (750 GB total RAM) · 180 app servers (100 RPS each, 50 threads @ 500ms)

RAM / Cache calculation

- Decide what to cache: last N posts per user, hot content, session data

- Cache size per user = N items x item_size_bytes
- Total RAM = cache_per_user x DAU
- Cache servers = total_RAM / RAM_per_server (typical: 64–128 GB)

App server calculation

- Threads per server: typically 50–100 for a web server
- Latency target: 500 ms = 2 requests/sec per thread
- Server capacity = threads x (1000ms / latency_ms)
- Servers needed = total_RPS / server_capacity

6. Steps 5 & 6: Latency and CAP Trade-off

Step 5 — Latency Estimation

State your latency target explicitly. For Facebook: **95% of requests served in 500 ms**. This p95 target directly drives the thread model and server count. Different latency targets (50ms, 100ms, 500ms) change the number of servers significantly.

Step 6 — CAP Theorem Trade-off

For Facebook, the recommended choice is **AP (Availability + Partition Tolerance)** — sacrifice Consistency. This means:

- System stays online even during network partitions or node failures
- Users may briefly see slightly stale data (e.g., like count off by 1)
- Data becomes eventually consistent once the partition heals
- Correct for social media — availability matters more than perfect consistency

> Banking/payments: choose CP — consistency is critical, downtime is acceptable.
>
> Social media/DNS/caches: choose AP — availability is critical, eventual consistency is fine.
>
> CA: not viable for distributed systems — a partition will always happen eventually.

8. Complete Summary Table

Complete Estimation Summary — Facebook

Parameter	Estimate	How calculated
Total Users	1 billion	Rounded for easy math
DAU	250 million (25%)	Assumed engagement rate
Queries/DAU/day	7 (5R + 2W)	User interaction frequency
RPS	~18,000	1.75B requests / 100K sec/day
Post size	500 bytes	250 chars x 2 bytes (Unicode)
Daily post storage	250 GB	250M x 1 KB (2 posts each)
Daily image storage	~8 TB	25M uploads x 300 KB
5-yr post storage	500 TB	250 GB/day x 2,000 days
5-yr image storage	16 PB	8 TB/day x 2,000 days
Cache RAM needed	750 GB	250M users x 3 KB (5 posts each)
Cache servers	10	750 GB / 75 GB per server
App servers	180	18,000 RPS / 100 RPS per server
Latency target	500 ms (p95)	95% requests served in 500 ms
CAP choice	AP (Availability+P)	Consistency sacrificed for resilience

Interview rule: BoE estimation takes max 10 minutes. Round aggressively. Justify every server and storage component.
Key formula: RPS = (DAU x queries/day) / 86,400 | Servers = RPS / (threads x RPS_per_thread) | Storage = users x data_size

All estimates for Facebook at scale — 1B users, 250M DAU, 18K RPS, 180 app servers, 10 cache servers

9. Key Formulas Cheat Sheet

RPS	DAU x queries/user/day / 86,400 (or / 100,000 for easy math)
Storage/day	DAU x posts/user x bytes/post + upload_rate x DAU x file_size
Storage total	daily_storage x 365 x years (use 2,000 days for 5 years)
Cache RAM	DAU x items_per_user x bytes_per_item
Cache servers	total_cache_RAM / RAM_per_server (typically 64 or 128 GB)
RPS per server	threads_per_server x (1000 / latency_ms)
App servers	total_RPS / RPS_per_server
Bandwidth (in)	RPS x avg_request_size_bytes
Bandwidth (out)	read_RPS x avg_response_size_bytes

Data Size Reference

1 char (ASCII) = 1 byte · 1 char (Unicode/UTF-8) = 2 bytes · 1 int = 4 bytes · 1 long/double = 8 bytes

1 KB = 1,000 B · 1 MB = 1,000 KB · 1 GB = 1,000 MB · 1 TB = 1,000 GB · 1 PB = 1,000 TB

Typical image = 200–300 KB · Thumbnail = 20–50 KB · Video (1 min HD) = ~50 MB · Audio (1 min) = ~1 MB

10. Quick Revision & Interview Questions

Quick Revision

- → BoE estimation comes BEFORE system design — justify every component with numbers.
- → Round aggressively: 1B users, 25% DAU, 7 queries/day — round to multiples of 10.
- → RPS formula: DAU x queries/day / 86,400 (use 100,000 for easy mental math).
- → Facebook: 250M DAU x 7 = 1.75B req/day / 100K = ~18,000 RPS.
- → Storage: estimate text (bytes) and images (KB) separately.
- → 5-year storage: multiply daily storage by 2,000 days (approx).
- → Facebook: 250 GB/day text → 500 TB over 5 years. 8 TB/day images → 16 PB over 5 years.
- → Cache RAM: DAU x items_per_user x bytes/item = 250M x 3KB = 750 GB.
- → Cache servers: 750 GB / 75 GB per server = 10 servers.
- → App servers: 1 server = 50 threads x 2 RPS = 100 RPS. 18,000 / 100 = 180 servers.
- → Latency target: p95 = 500 ms. Thread can handle 2 req/sec at 500 ms.
- → CAP for Facebook: AP — availability + partition tolerance, eventual consistency.
- → Take max 10 minutes on BoE in interviews. State assumptions explicitly.
- → Unicode chars = 2 bytes. ASCII = 1 byte. int = 4 bytes. long/double = 8 bytes.

Interview Questions

- ■ 1. Walk me through a back-of-the-envelope estimation for a system with 1M DAU.
- ■ 2. How many requests per second would Facebook handle with 250M DAU?
- ■ 3. How would you estimate storage for a social media platform over 5 years?
- ■ 4. How many application servers are needed for 18,000 RPS?
- ■ 5. How would you estimate RAM needed for a caching layer?
- ■ 6. What is p95 latency and how does it affect server count estimation?
- ■ 7. What CAP trade-off would you choose for a social media platform? Why?
- ■ 8. Why do we divide by 86,400 when calculating RPS?
- ■ 9. How do you estimate bandwidth requirements for a system?
- ■ 10. What assumptions would you make for a YouTube-scale storage estimation?

Key Terms

BoE Estimation	Rough calculation before design to estimate servers, storage, RAM, bandwidth
DAU	Daily Active Users — users who interact with the system on a given day
RPS	Requests Per Second — total throughput the system must handle at peak
p95 Latency	95th percentile latency — 95% of requests complete within this time
Storage horizon	5 years is the standard interview assumption (approx 2,000 days)
Thread model	Server handles N threads; each thread processes 1 req per latency window
Cache hit rate	Fraction of reads served from cache without hitting DB (typically 80–90%)
AP system	Chooses Availability + Partition Tolerance — eventually consistent

CP system	Chooses Consistency + Partition Tolerance — may be unavailable on partition
Unicode char	2 bytes per character — use when storing multi-language text
T-shirt sizing	Using round numbers (1B, 10M, 100K) for quick mental math in estimation